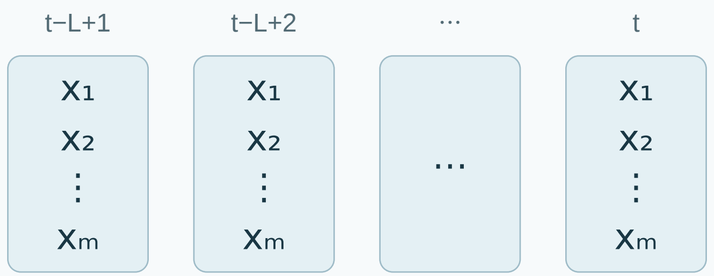
Parameter-conditioned SHRED prediction

One complete sensing window + a candidate rotation → a future pressure field

1 / ASSEMBLE THE INPUT

L timesteps of sparse measurements



Add the candidate rotation as a feature

X_t: L samples × m pressure sensors

Ω_j: one added rotation channel

$[X_t \mid \Omega_j]$

2 / SHRED SURGROGATE

One trained model

Recurrent encoder
LSTM · temporal history

↓

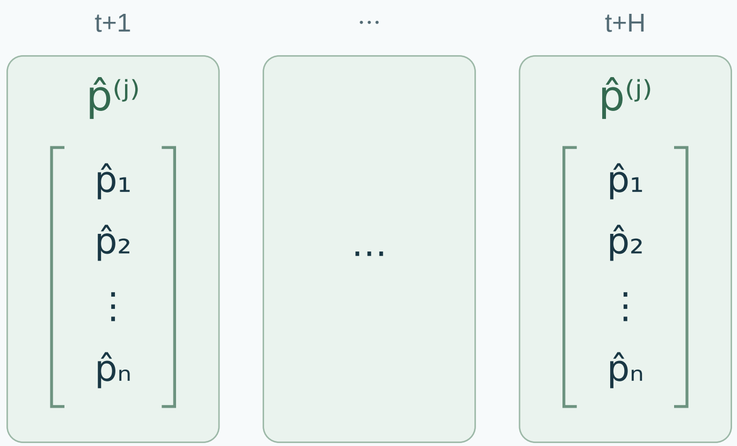
Shallow decoder
+ spatial reconstruction

Same weights for
every candidate rotation

predict

3 / DECODE THE PREDICTION WINDOW

Full pressure fields for candidate ω_j



n spatial locations · H future timesteps

A predicted field on the same mesh

VARY THE PARAMETER · KEEP THE SENSING WINDOW AND MODEL FIXED

For each candidate value, repeat the same prediction:

$\omega_1 \rightarrow \hat{p}^{(1)}$

$\omega_2 \rightarrow \hat{p}^{(2)}$

Different rotation values give different predicted pressure fields on the same spatial domain.

Conceptual constant-parameter example. MPC uses ramp-and-hold rotation profiles and the configured observation–prediction time offset.